

Tutorial - Week 08 5COSC019W – Object Oriented Programming – Java

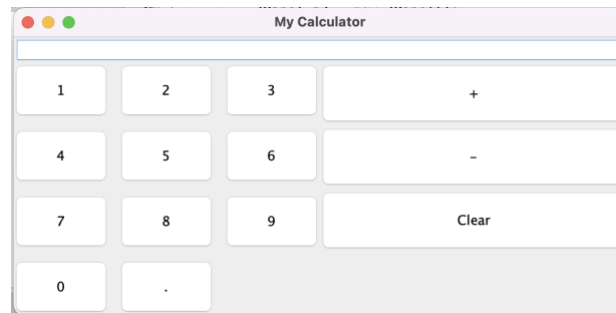
Implement a GUI App and a JTable

10 - 11 November

GUI application

1) Write a Swing GUI application called “My Calculator”, as shown in the figure below.

In this exercise, you need to use a combination of Layouts to arrange the components as shown.

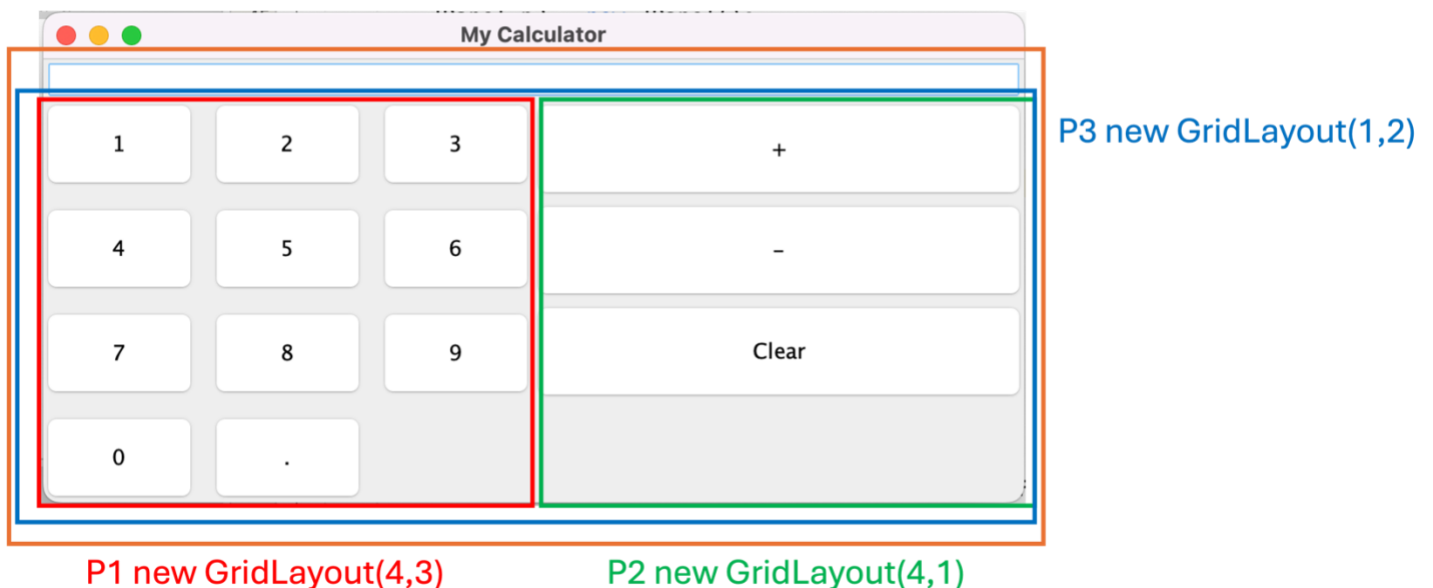


For some Guidance (please refer also to the “Microwave” exercise in the lecture week 07). The user interface has:

- JButton
- JTextField

They are organised using different panels with different layouts.

BorderLayout (JTextField in the North and P3 in the CENTER)



Note: Swing Components are kept in the package javax.swing. They begin with a prefix "J", e.g., JButton, JLabel, JFrame.

```
import java.awt.*;           // Using AWT's layouts
import javax.swing.*;        // Using Swing components and containers
```

Write a class CalculatorFrame that extends JFrame:

```
public class CalculatorFrame extends JFrame{

    // Constructor to setup UI componentspublic CalculatorFrame () {
        Your code here for the constructor

    // set here the layout for the different panels
    Your code here is for creating the panels, setting the layout, and adding the
    components.

}
```

In the main method, an instance of the CalculatorFrame is created and made visible:

```
public static void main(String[] args) {
    // TODO code application logic here
    CalculatorFrame myCalculator = new CalculatorFrame();
    myCalculator.setVisible(true);
    myCalculator.setSize(600, 300);
    myCalculator.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}
```

2) JTable

Download the following NetBeans project from BB Tutorial Week 08, or from the following link:

[PersonProject.zip](#)

In this Java Project, you have the following classes::

- The **Person** class, is the superclass for Teacher and Student and holds information about the *name* and the *date of birth* of a person.
- The **Teacher** class is a subclass of Person and should include appropriate methods and hold information about the *salary*.
- The **Student** class is a subclass of Person and should include methods and information about *ID* and the *Course* they are studying.
- The class **Date** is used to represent the *date of birth* of each person.
- The class **PersonProject** has the main() method and here some teachers and students have been created.

TASK:

- 1) Display the information relative to each person in a frame, through a JTable. In particular, you should display the name, the date of birth and the type (if it is a teacher or a student) in each column of the table.

Hint 1:

Create a custom TableModel called PersonTableModel that holds the data and extends the AbstractTableModel. Here you can see the implementation. Try first by yourself and then check the following:

```
public class PersonTableModel extends AbstractTableModel{

    private String[] columnNames = {"Name", "Date of Birth", "Type"};
    private ArrayList<Person> list;

    public PersonTableModel(ArrayList<Person> personList){
        list = personList
    }
}
```

```
public int getRowCount() {
    return list.size();
}
```

Override the abstract methods of
AbstractTableModel

@Override

```
public int getColumnCount() {
    return columnNames.length;
}
```

@Override

```
public Object getValueAt(int rowIndex, int columnIndex) {
    Object temp = null;
    if (columnIndex == 0) {
        temp = list.get(rowIndex).getName();
    }
    else if (columnIndex == 1) {
        temp = list.get(rowIndex).getDOB().getDate();
    }
    else if (columnIndex == 2) {
        if (list.get(rowIndex) instanceof Teacher)
            temp = "Teacher";
        else
            temp = "Student";
    }
    return temp;
}
```

Check the instance type and set the value to be Teacher or Student

Hint 2:

Let's now create a class PersonTable, representing our frame where the table is displayed.

Below, you find the main structure of this class, with comments on the part you need to add.

```
public class PersonTable extends JFrame{

    JTable myTable;
    PersonTableModel tableModel;
    ArrayList<Person> list;

    // constructor
    public PersonTable(ArrayList<Person> list){

        // initialise and instantiate the instance variable
        // create an instance of PersonTableModel
        // create an instance of JTable and link the model to the table

        // set the size of the frame

        // instantiate a JScrollPane and add the table BorderLayout as default
    }
}
```

Hint 3:

Create an instance of your frame, called PersonTable, in your main() method and make the frame visible.

If you run your project, you now should see your table on the screen.

If you are confident in writing Java and you finish these tasks, independently, you can attempt the following advanced exercises:

- 1) Sort the column of a JTable using a row sorter.
Hints: from the Java API investigate how to use the method `setAutoCreateRowSorter` from the JTable class.
- 2) Colour a table cell of the type or person as follows: green if it's a teacher, blue if it's a student.
Hint: Investigate the use of `getTableCellRendererComponent`

Solution will be provided